Customer Shopping Behavior Analysis

This project analyzes customer shopping behavior using transactional data to uncover insights into spending patterns, customer segments, and product preferences. Our goal is to guide strategic business decisions.



Project Overview & Dataset Summary

Project Goal

Uncover insights from 3,900 purchases to inform strategic business decisions regarding spending patterns, customer segments, product preferences, and subscription behavior.

Dataset at a Glance

- Rows: 3,900
- Columns: 18
- Key Features: Demographics, Purchase Details, Shopping Behavior
- Missing Data: 37 values in Review Rating

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Exploratory Data Analysis with Python

01

Data Loading & Exploration

Imported dataset using pandas; used `df.info()` and `.describe()` for initial insights.

02

Data Cleaning

Handled missing values in 'Review Rating' by imputing with median rating per category.

03

Column Standardization

Renamed columns to snake case for improved readability and documentation.

04

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns.

05

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

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Analysis Using MS SQL Server

Subscription

Subscribers vs. Non-Subscribers

Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

Discount Dynamics

Discount-Dependent Products

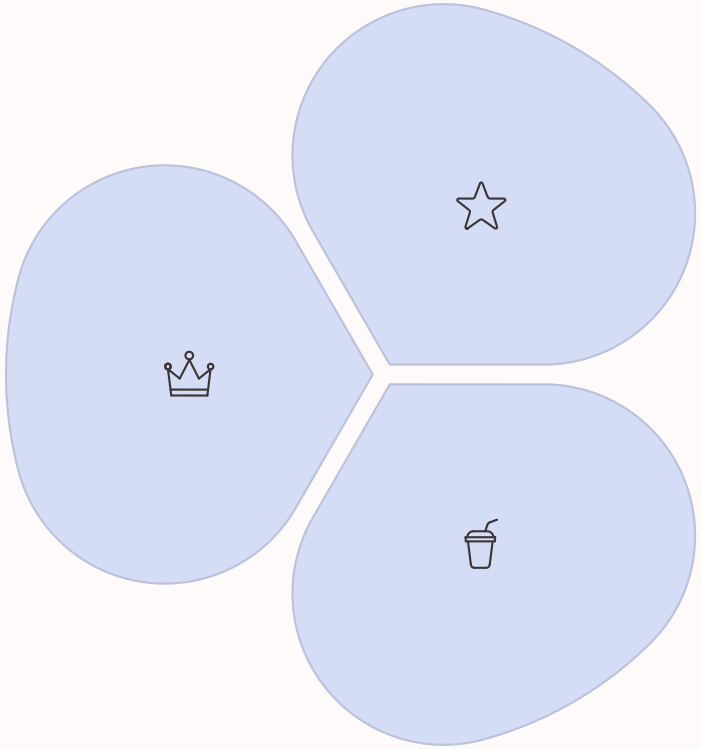
Hat, Sneakers, Coat, Sweater, and Pants had the highest percentage of discounted purchases, indicating potential discount dependency.

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Customer Segmentation & Product Preferences

Loyal

3,116 customers



New

83 customers

Returning

701 customers

Customers were classified into Loyal, New, and Returning segments based on purchase history.

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Top Products by Category

1	**Accessories** • Jewelry (171 orders) • Sunglasses (161 orders) • Belt (161 orders)
2	**Clothing** • Blouse (171 orders) • Pants (171 orders) • Shirt (169 orders)
3	**Footwear** • Sandals (160 orders) • Shoes (150 orders) • Sneakers (145 orders)
4	**Outerwear** • Jacket (163 orders) • Coat (161 orders)

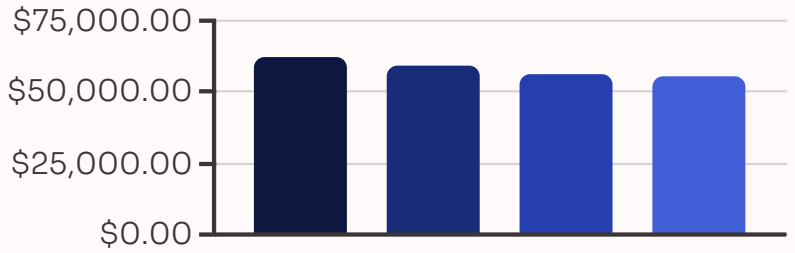
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Repeat Buyers & Revenue by Age Group

Repeat Buyers & Subscriptions

Customers with more than 5 purchases are more likely to subscribe (958 subscribers vs. 2518 non-subscribers).

Revenue by Age Group



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Power BI Dashboard:Visualizing Insights



An interactive Power BI dashboard provides a comprehensive visual overview of customer behavior, including key metrics like average purchase amount ($0.1K), average review rating (3.75), and customer count (3.9K).

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Strategic Business Recommendations

Boost Subscriptions

Promote exclusive benefits to increase subscriber base.

Customer Loyalty Programs

Reward repeat buyers to foster loyalty and move them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control for discount-dependent products.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users for maximum impact.

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